Topic 4: Terraform Blocks

1. Provider Block

Providers tell Terraform which cloud service to use and how to authenticate. These blocks define the plugin source and versioning constraints.

```
terraform {
  required_providers {
    aws = {
      source  = "hashicorp/aws"
      version = "~> 5.0"
    }
  }
}

provider "aws" {
  region = "us-east-1"
  default_tags {
    tags = { ManagedBy = "Terraform" }
  }
}
```

2. Variable Block

Input variables make code reusable and dynamic.

- **Types:** string, number, bool, list(number), map(string), object({...})
- **Validation:** Use `condition` blocks to enforce input standards.
- **Sensitivity:** Use `sensitive = true` to obscure values in terminal outputs.

```
variable "environment" {
  type        = string
  description = "dev/staging/prod"
  validation {
    condition     = contains(["dev", "staging", "prod"], var.environment)
    error_message = "Environment must be dev, staging, or prod."
```

```
  }
}
```

3. Output Block

Outputs expose important infrastructure values, functioning like return values in a function.

```
output "instance_public_ip" {
  description = "Public IP address of the web server"
  value       = aws_instance.web.public_ip
}
```

4. Resource Block

Resources are the atomic unit of infrastructure. Terraform automatically generates a dependency graph based on the references within these blocks.

```
resource "aws_instance" "web_server" {
  ami           = "ami-0abcdef1234567890"
  instance_type = "t3.micro"
  subnet_id     = aws_subnet.public.id

  root_block_device {
    volume_size = 20
  }
}
```